

System-Level PPA Trade-Off Matrix

Metric	Static CMOS	Transmission Gate
Area	466 μm^2	288 μm^2 (WINNER)
Power	2.45 μW	1.72 μW (WINNER)
Post-Layout Delay	1.77 ns (WINNER)	1.97 ns
Efficiency (PDP)	4.35 fJ	3.36 fJ (WINNER)

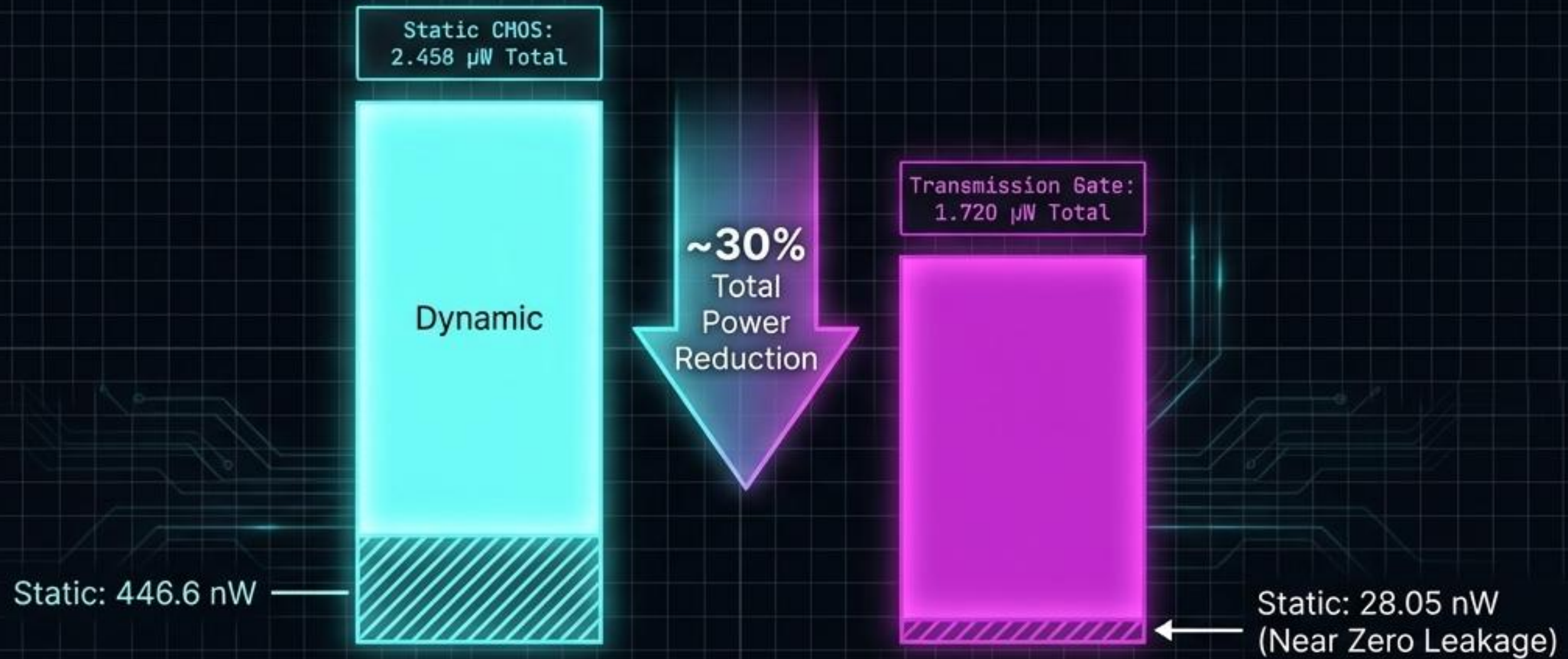
Transmission Gate architectures offer massive area and efficiency gains, but Static CMOS remains mandatory for parasitic-heavy critical timing paths.

Physical Footprint: Area Optimization



Transmission Gate architecture reduces overall transistor count and simplifies routing complexity. This yields a highly compact footprint, ideal for area-constrained layouts.

Electrical Efficiency: Power Reduction



Lower capacitance and fewer switching transistors drive down dynamic power. Simultaneously, the TG topology nearly eradicates static leakage, resulting in dramatically lower total power consumption.

The Post-Layout Reality Check

Ideal Simulation (Pre-Layout)

"TG outperforms CMOS."

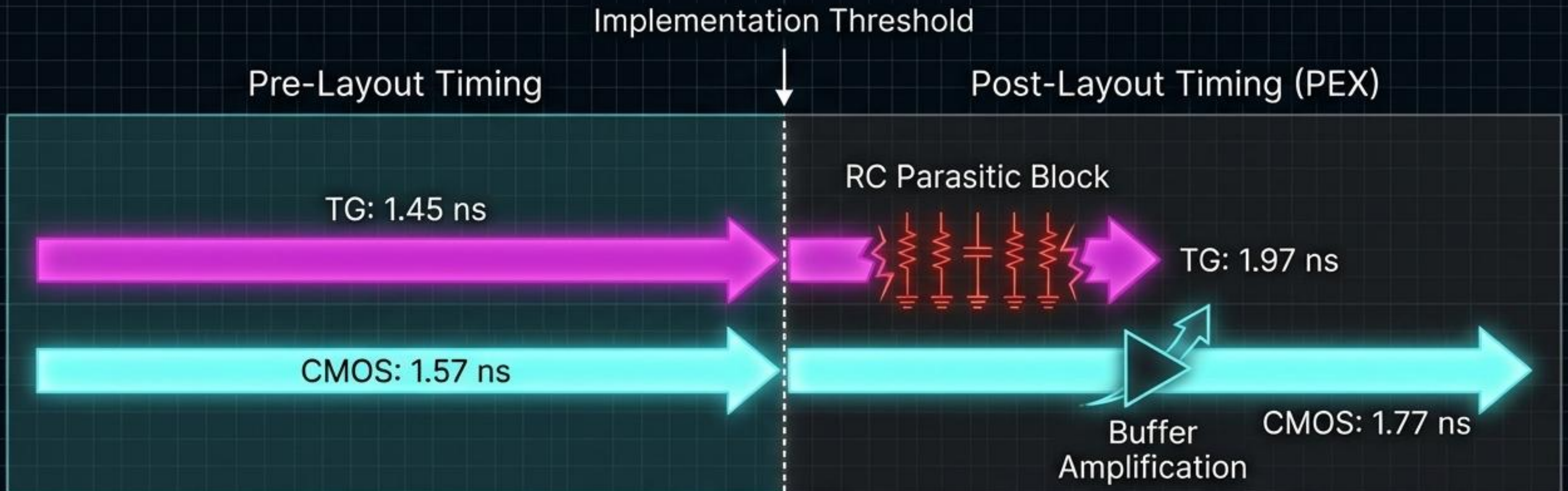
Physical Reality (Post-Layout PEX)

"Routing parasitics degrade TG performance."

Core Insight

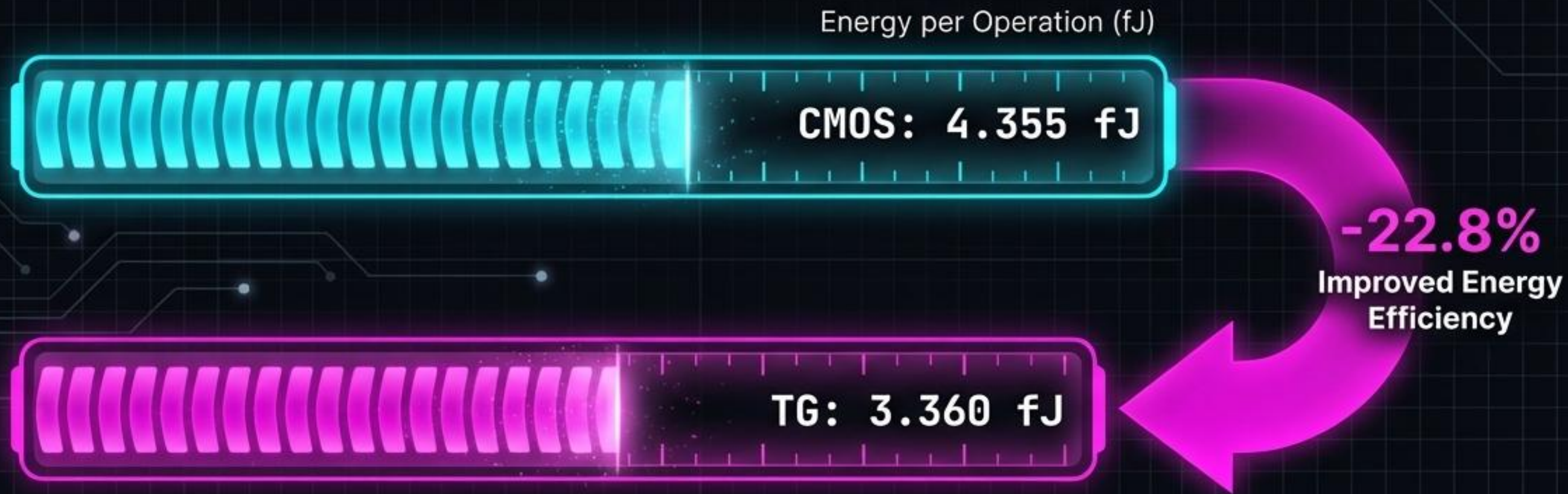
Unlike standard CMOS buffers that actively restore signal strength, Transmission Gates act as passive RC delay lines, making them highly sensitive to post-layout parasitic loads.

Performance: The Delay Reversal



Pre-Layout: Transmission Gate executes faster.
Post-Layout: RC Parasitic effects heavily penalize the TG signal path, shifting the speed advantage back to buffered Static CMOS.

The Ultimate Metric: Power-Delay Product (PDP)



Despite the post-layout delay penalty, the massive reduction in power consumption compensates for the slower timing. Transmission Gate remains the superior choice for overall energy efficiency (PDP).